MINOR PROJECT 5: NYC PROPERTY SALE PRICE PREDICTION

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Objective

The project builds and compares regression models to estimate New York City property sale prices from location and building characteristics. **Regression** is a machine-learning task used to predict a continuous numerical value, such as a property price.

Dataset Overview

The CSV contains **34,439 rows and 19 columns**, with each row representing a property sale. The target column is **sale_price**. Input information includes borough, latitude and longitude, year built/building age, lot and building area, residential and commercial area, unit counts, floors, land use, and building class. The dataset initially contained missing values and 1,003 duplicate rows.

Work Completed

• **Exploration:** checked dimensions, data types, descriptive statistics, missing values, and duplicates.
• **Cleaning:** removed duplicates, rows with inconsistent borough labels, zero building area/units, and sales below $10,000; filled missing area/floor values with medians and land use with its mode. The cleaned dataset contained **24,591 rows**.
• **Feature preparation:** grouped building-class labels by their first letter, treated land use as categorical, removed ID/redundant columns, applied one-hot encoding, and transformed sale price using **log1p** to reduce right-skewness.
• **Exploratory analysis:** plotted price distributions, median prices by borough and building-age group, building area versus price, and a numerical correlation heatmap.
• **Training:** split data into 80% training (19,672 rows) and 20% testing (4,919 rows), then trained Linear Regression, AdaBoost, Gradient Boosting, and XGBoost models.

Model Results

Model	R^2 (log price)	RMSE (log price)	MAE (USD)
XGBoost	0.6101	0.4917	$345,026
Gradient Boosting	0.5857	0.5068	$358,460
Linear Regression	0.3598	0.6300	$499,941
AdaBoost	0.1562	0.7233	$587,908

Definitions of Evaluation Metrics

R^2 (coefficient of determination): indicates how well predictions explain variation in the target; higher is generally better.
RMSE (root mean squared error): measures prediction error while penalizing larger errors more strongly; lower is better. Here it is calculated on log-transformed prices.
MAE (mean absolute error): the average absolute difference between actual and predicted prices in dollars; lower is better.
Boosting: an ensemble method that builds models sequentially, with later trees learning from earlier errors. **Feature importance** estimates how much each input contributes to the trained tree model.

Conclusion

XGBoost performed best among the four tested models (R^2 = 0.6101; MAE ≈ $345,026). The notebook's feature-importance analysis ranked the Manhattan borough indicator highest, followed by residential-unit count and land-use indicators. Results show useful predictive signal, but the remaining error is substantial; the dataset lacks details such as individual-unit condition/bedrooms and sale dates, which may limit accuracy.